1 Research question

Does the concentration of young people leaving school at the same time affect crime in the hours after release?

Reduced form. Did the 2014 change in the schedule of 1st-year EM classes, which stopped them from leaving school at the same time as the 2nd and 3rd years, change crime by or against 2nd- and 3rd-year students after release, in schools that used to release all EM years together?

2 Motivation

School reduces youth crime in two ways: it raises human capital in the long run (Lochner and Moretti, 2004), and it keeps students in a supervised environment, where they can not commit crime (incapacitation). But school also brings young people together, and that can increase crime. Billings et al. (2019) use the redrawing of school boundaries in Charlotte-Mecklenburg in 2002 and find that one standard deviation more same-school, same-grade, same-race and same-gender peers living within 1 km raises the probability of arrest at ages 16–21 by 3.9 percentage points (23%). Two students who live within 1 km of each other are about six times more likely to be arrested *together* if they are assigned to the same school.

Concentration can also raise crime in the short run, while students are together. The literature shows evidence of the concentration effects through variation in session-days and the type of crime. Jacob and Lefgren (2003) show that propery crime falls by 14% on school days, but juvenile violent crime rises by 28%, and the violent crime peaks at about 3 p.m., when school lets out. The authors arugue that the fall in property crime is an evidence of incapacitation effects, and the rise of violent crime is an effect of concentration. Luallen (2006) follows the same line and shows that the property crime increases +29% and violent crime −32% on strike days.

This is weak evidence of a concentration effect for some reasons. First, they identify only the net effect of incapacitation and concentration. The effect of school on violent crime is the sum of incapacitation (−) and concentration (+). Since the net effect is positive (+28%), concentration outweighs incapacitation for violent crime. The effect on property crime is the sum of incapacitation (−) and concentration, whose sign is unknown. If we assume it is non-negative, the negative net effect (−14%) means that incapacitation outweighs concentration for property crime. Under these sign assumptions, they identify only the lower bound of the concentration effect.

Second, the source of variation is a whole day without school, which changes substantially what students do during the day, not only how many other students they meet. The changes in property and violent crime can therefore reflect the different opportunities students face on an "atypical" day: the same student can change which crime they commit. For example, a student who would have fought at school (violent crime) may instead shoplift at a mall (property crime), because a free day leaves time to comute to the mall that is far away. If students substitute across crimes in this way, the type of crime does not reveal whether incapacitation or concentration is at work. This fits the results of Jacob and Lefgren (2003): on in-service days, total serious juvenile crime is almost unchanged, because the rise in property crime and the fall in violent crime roughly cancel out.

Third (and less severe), although Jacob and Lefgren (2003) rules out the bias of school-reporting by estimating the effect of non school hours crimes or crimes that occured out of schools, their effect decrease with their significance.

Separating concentration from incapacitation is hard. Any policy that changes when students leave school usually also changes how long they stay there, and so how much time they spend

supervised. This paper uses a source of variation that changes concentration while keeping school and supervision fixed: whether 9th graders are released at the same time as high-school students in their own school. We compare classes that have school on the same day and spend the same number of hours in class. Only the timing of release differs, and with it the number and the age of the students who leave with them. This design has four advantages over the existing evidence:

1. **Incapacitation is held fixed.** School is in session on both sides of the comparison, and class duration is the same. Treated and control students have the same hours of supervision and the same free time. Any difference in crime comes from who they are released with, not from how long they were in school.

2. **Concentration is measured, not inferred.** We observe how many students leave the school at the same time, and within any window around the bell. This gives a dose–response: crime as a function of the size of the crowd at the gate, not just the sign of an effect.

3. **Size and composition can be separated.** Sharing release time with high-school students raises both the number of youths on the street and their age. Comparing exposure to older students with exposure to the same number of same-age students separates a pure concentration effect from a peer effect of older students.

4. **It targets the right margin.** The effect is measured on the street after release, outside school grounds, where reporting by school staff plays no role. That is the margin a change in bell times acts on.

The setting adds external value. Most of the evidence comes from US cities. In Brazil, public schools run several shifts per day, so release times vary a lot within and across schools. Youth violence is also much higher, so the stakes of getting the policy right are larger.

Understanding the contemporaneous effect of concentration matters for two reasons:

1. It is cheap to prevent. If shared release raises crime, staggering bell times is a low-cost policy, unlike lengthening the school day or reassigning students across schools.

2. It may feed the long-run effect. If crimes committed together after school are where partnerships start, the short-run concentration effect is one channel behind the networks documented by Billings et al. (2019). Luallen (2006) finds that juveniles who become repeat offenders are more likely to commit their first crime on a strike day.

We also investigate:

1. **Displacement.** Does staggered release reduce crime, or only shift it to other times and days? Jacob and Lefgren (2003) and Luallen (2006) find no temporal displacement of violent crime (leads and lags, and weekly and monthly totals). We can run the same tests.

2. **Place.** Where does concentration-driven crime happen: at the school gate, or along the routes between school and home?

3. **Type of crime.** Is the effect driven by fights between students (assaults with young victims) or by property crime? Is it larger for crimes with several juvenile offenders? Jacob and Lefgren (2003) find no evidence that school helps juveniles coordinate crimes. Billings et al. (2019) find that partnerships are common among older youth.

4. Crime against students

3 Mechanisms

I can think of three mechanisms on which conecetration of youth can increase crime. First, is the numeric concentration of people, second the peer effect and third the report. Each of these mechanisms operate in the different manner.

A. First, the number of people: how many young people are around. A concentration of people can increase crime because:

   (a) there are more interactions, which can help youths coordinate crimes. Jacob and Lefgren (2003) find no evidence that school increases the coordination of crime: crimes with several offenders do not respond more to school days. We can test whether the number of crimes committed by more than one student increases.

   (b) there are more people who can fight each other. We can test this, but we cannot separate it from old rivalries (the peer effect below).

   (c) there is less adult supervision per youth (police, school staff, security guards on the street), as long as the police do not respond to the crowd.

   (d) more students can attract offenders, which increases crime against students. We can check whether the number of student victims increases.

B. Second, the peer effect: .

   (a) **Learning.** Students can learn from other students. In our main specification, we are measuring the exposition of older students to younger ones so this channel may be weaker in our case. But we can test (long-shot) if there is an increase in crimes committed jointly with a prior offernder and a first-time offender. Perhaps we do not have enough variation.

   (b) **Recruitment.** If a 3rd-year student pulls 1st-year students into a theft, the 3rd-year student's own crime also increases, because they offend together. We can test this with crimes committed by mixed-grade groups.

   (c) **Status.** Older students may act out/show off to impress younger ones.

   (d) **Within student rivalries.** Students can fight because of their rivalries, including hazing of 1st-year students by older students. This predicts fights across grades, with older offenders and younger victims. We can test this.

C. Third, policing and reporting (I am not sure if we treat this as a mechanism or a confounder...)

   - **Reporting.** A crowd can change the share of crimes that reach the police, in either direction. It can decrease reporting: offenders are harder to identify in a crowd and each witness expects someone else to call Also, fights between students that would otherwise happen away from the school may happen at the school gate when students are released together. There, the school is more likely to handle them as a discipline matter, so they are less likely to be reported to the police. It can increase reporting: there are more witnesses and more phones to film or call.
   - **More police.** When students leave together, the police can respond with more patrols at the school gate. This has two effects: more police can deter crime (a real change in behavior), and more police can detect more,

We can test for these channels by checking whether, with shared release:

(a) the average time between the crime and the arrival of the police (or the report) changes;

(b) the share of crimes in which the police are already at the scene, or arrest the offender in the act (*flagrante*), increases;

(c) the share of incidents started by the police (patrol) rather than by a call from a victim or witness changes;

(d) serious crimes, which are reported whoever is watching , respond in the same way as minor crimes. If only minor crimes respond, the effect is more likely to come from reporting.

4 Population of Study

- **Population.** Students in the 2nd and 3rd years of high school (EM) in state schools in Minas Gerais that offer all three years of EM.
- **Treatment.** The class is released at the same time as the 1st-year EM classes of the same school and shift. A continuous version is the number of 1st-year students released within X minutes of the class.
- **Outcome.** Crime committed by (and against) 2nd- and 3rd-year students after release, near the school and in the minutes after the bell.

5 Identification Strategy

5.1 Motivation for the Identification Strategy

The naive comparison. The simplest design compares schools that release the three EM years together with schools that release each year at a different time. But release times are not random, and this comparison is biased by confounders at three levels.

School level.

- **Size and density.** Schools with fewer classes can release everyone together, so shared release is more common in smaller schools. Smaller schools are in less dense areas with less crime. This biases a comparison of shared and non-shared schools towards finding that concentration *reduces* crime. The bias reverses if treatment is measured as the *number* of students released together, which is larger in large schools located in dense areas.
- **Infrastructure.** Staggered times often come from lack of rooms or extra shifts, which are also related to school quality.
- **Reverse causality.** Principals may stagger release *because* of fights at the gate, so treatment responds to past violence.
- **Nearby schools.** Other schools releasing at the same time add to concentration, and their students may also be affected.

Individual level.

- **Sorting into schools.** Families choose schools, so students in small schools with shared release differ from students in large schools (income, distance from home, prior behavior).
- **Sorting into shifts and classes.** Schools assign students to shifts and classes by age, repetition, behavior or work status. If some classes are released at different times, the students in them may differ in their risk of crime.

Time of day. A different release time is a different clock time, with different parental supervision, daylight, policing and street activity.

A source of variation: the 2014 change in 1st-year schedules. In 2014, 1st-year EM classes in state schools in Minas Gerais became about 50 minutes longer, for that year only (probably the sixth period of *Reinventando o Ensino Médio*; to be confirmed). Morning 1st-year classes moved their release from 11:20–11:30 to about 12:10–12:20, and afternoon classes from 17:00–17:30 to 17:50–18:20. Start times did not change, and the schedule of 2nd-year classes did not change either: they kept leaving at 11:20–11:30 and 17:00–17:30. In 2015 the 1st year went back to its previous schedule.

For a 2nd- or 3rd-year student in a school that released all EM years together, this means:

- **Concentration falls in 2014 and returns in 2015.** The 1st-year students no longer leave with them in 2014, and leave with them again in 2015.
- **Their own schedule is unchanged.** Start time, class duration and release time are the same, so incapacitation (hours in school, free time) and clock time are held fixed *for the treated students*.
- **The change is set by the state, not by the school.** It does not respond to violence at a particular school's gate.

Design. Compare 2nd- and 3rd-year students in schools that shared release with the 1st year before 2014 (treated: sharing stops in 2014 and returns in 2015) with students in schools that did not share release (control: no change), in 2012–2016. The reversal in 2015 is a second test: if crime falls in 2014 because of lower concentration, it should rise again in 2015.

- **Comparison:** treated vs control schools, before, during and after 2014.
- **Unit:** 2nd- and 3rd-year class (or school $\times$ shift) $\times$ year.
- **Fixed effects:** school $\times$ shift and year.
- **Identifying assumption:** without the 2014 change, crime around the release of 2nd- and 3rd-year students would have evolved in parallel in treated and control schools; and the 2014 change affected these students only through the number of students released with them.

Threats.

- **Treated and control schools differ.** Schools that shared release are smaller (see above). Fixed effects absorb level differences, but not different trends. Check pre-trends in 2010–2013.
- **The 1st-year students also change.** In 2014 they leave later and spend more time in school, so their own crime changes. The outcome must be crime by (or against) 2nd- and 3rd-year students, not all youth crime near the school.

- **Other changes in 2014.** The policy may have come with other changes at the school (activities, staff) that affect all students. In addition, Belo Horizonte hosted World Cup matches in June–July 2014, with changes to the school calendar and heavy policing, and 2014 was an election year. Year fixed effects absorb what is common to all schools, but not effects that differ between treated and control schools.
- **The 3rd year is not yet checked.** The unchanged schedule is confirmed for the 2nd year (etapas 26/31/36). The 3rd year (etapas 27/32/37) still needs to be verified.
- **Compliance.** Not all schools may have changed the 1st-year schedule in 2014. Use the observed change in each school's schedule, not just the year, and count how many schools actually lost shared release between the EM years.

Descriptive evidence on the 2014 change. Figure 1 counts the schools whose main 1st-year morning release time moved 50 minutes or more later than in the previous year. The change is concentrated in 2014: 170 of 247 schools (69%). A few schools made the same change in earlier years: 8 from 2011 to 2012 and 11 from 2012 to 2013; they may have adopted the change before the rest of the network. These 19 schools are dropped from Figures 2 and 3, so the remaining schools are all untreated before 2014.

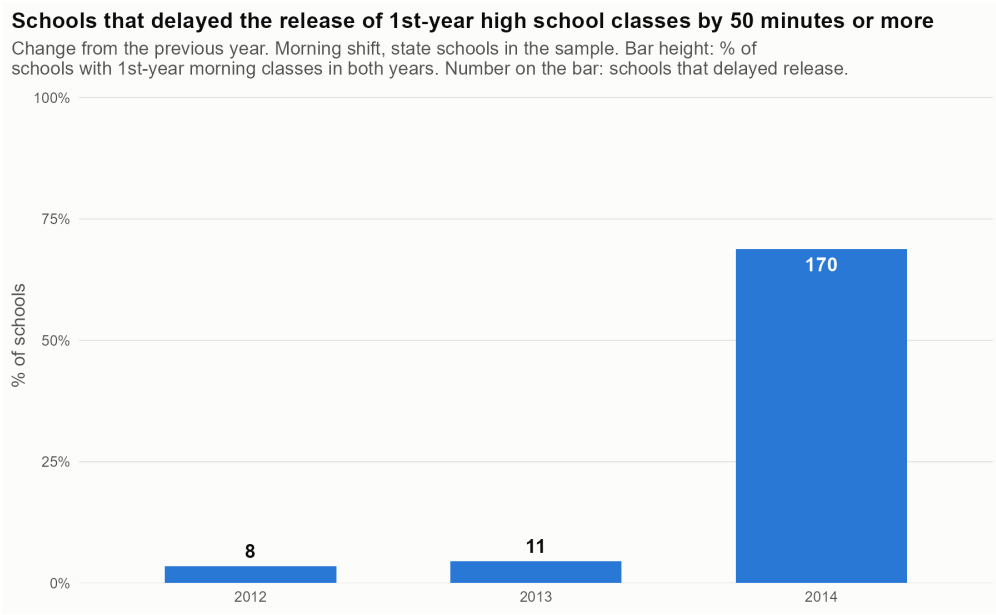


Figure 1: Schools whose main 1st-year EM release time in the morning shift moved 50 minutes or more later than in the previous year. Bar height: share of the sample schools with 1st-year morning classes in both years; number on the bar: number of schools.

Figure 2 shows that, in the morning, 1st-year classes moved from 11:20–11:30 to 12:15–12:20 in 2014 only, while 2nd-year classes kept leaving at 11:20–11:30. Figure 3 shows that, in the afternoon, some schools moved the start time earlier in 2014 (from 13:00 to 12:30). This happens for the 2nd year as well as the 1st, so in the afternoon the 2014 change does not leave the 2nd-year schedule unchanged in every school.

Figure 4 compares, within each school, the release time of the 1st year with that of the 2nd and 3rd years in the morning shift. In 2013 and 2015 almost every school releases all EM years at the

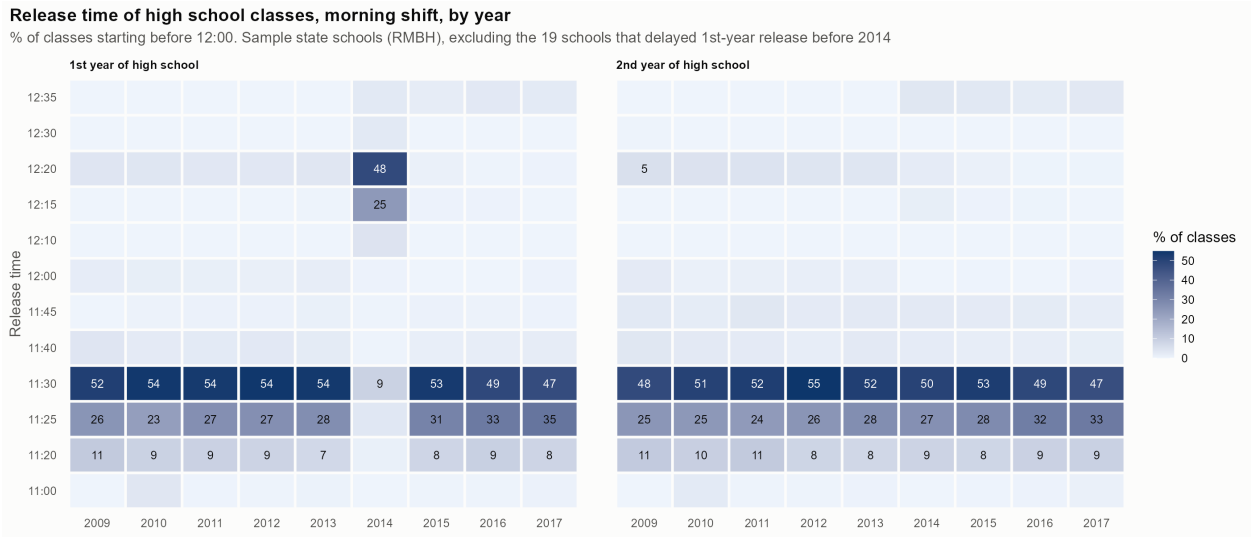


Figure 2: Release time of 1st- and 2nd-year EM classes in the morning shift (classes starting before 12:00), % of classes by year. Sample state schools, excluding the 19 schools that delayed 1st-year release before 2014.

same time (difference of zero). In 2014 most schools release the 1st year about 50 minutes after the 2nd and 3rd years: the median difference is 50 minutes, and 163 of 224 schools have a difference of 40 minutes or more. Only 37 schools keep releasing all years together in 2014.

I have educational data for MG from 2007 to 2017, including proficiency scores

5.2 Heterogenous Estimation

Powered effect of concentration: if have two shifts in the smae school. The program makes students go earlier in the school in the afternoon shift (starts at _)

5.3 Details:

- Schools with distinct release time of the class within the grade (around 5%): I would exclude them first, and then think
- Windows of the outocme: 1,2 ,3 hours
- Effect of morning shift and afternoon shift

5.4 Sources of variations

(a) "RCT" within school, class level.
Level. Class **Treatment.** Shared release time with 10th grade **Sample.** Cell = school × shift × year × class duration. Cells with at least two 9th-grade classes, at least one released at a time shared with 10th grade and at least one released at a time not shared with 10th grade. Treated and control classes have the same duration.
Estimator. Simple difference in means:

$$\hat{\beta} = \sum_s w_s \left(\overline{V}_s^{\text{shared}} - \overline{V}_s^{\text{not shared}} \right) \tag{1}$$

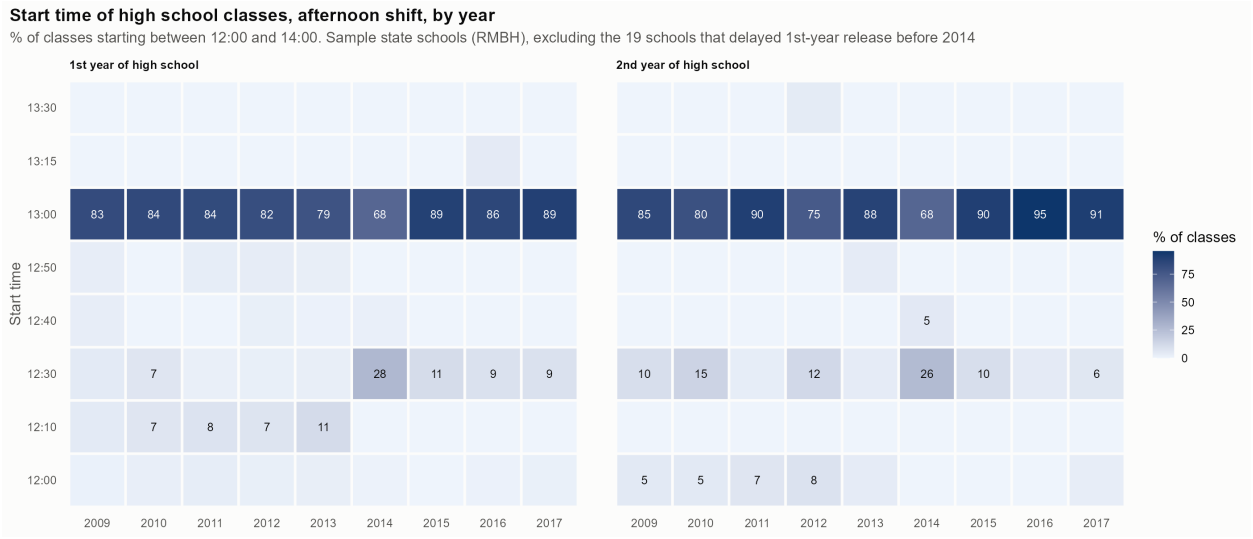


Figure 3: Start time of 1st- and 2nd-year EM classes in the afternoon shift (classes starting between 12:00 and 14:00), % of classes by year. Sample state schools, excluding the 19 schools that delayed 1st-year release before 2014.

Equivalent regression, with α_s a cell fixed effect:

$$V_{cs} = \alpha_s + \beta \, \text{Shared}_{cs} + \varepsilon_{cs} \tag{2}$$

Identifying assumption. Inside a cell, which class gets the shared release time is as good as random: students in shared and not shared classes of the same school, shift and year have the same potential outcomes. School-level confounders (size, density, neighborhood) and shift selection are fixed by construction.

Confounders.

- Confounder: the school chooses which class goes to which time (e.g. older or more "difficult" students placed together).
- Confounder: with the same class duration, a different release time means a different start time, so clock time (start and release) still differs between treated and control. Time in school does not.

(b) TWFE — between schools & within schools across time, not-yet-treated control. Level. School. **Treatment.** school s goes from some shared release time with high school to zero shared release time. Control: school m still has some shared release time in both periods (not-yet-treated). Restriction: the shift of the 9th grade does not change between $t-1$ and t; the shift of high school can change.

$$\hat{\beta}_{DD} = \left(\overline{V}_{s,t} - \overline{V}_{s,t-1}\right) - \left(\overline{V}_{m,t} - \overline{V}_{m,t-1}\right) \tag{3}$$

$\overline{V}_{s,t}$ = average violence among 9th-grade students at school s in year t.

- Confounder: change in class-time attract more violent students for 9th grade.

(c) Dif-dif, school-shift level. Level. School × shift (morning/afternoon).

$$\hat{\beta}_{DD} = \left[\overline{V}^{\text{morning}}_{\text{shared}} - \overline{V}^{\text{morning}}_{\text{not shared}}\right] - \left[\overline{V}^{\text{afternoon}}_{\text{shared}} - \overline{V}^{\text{afternoon}}_{\text{not shared}}\right] \tag{4}$$

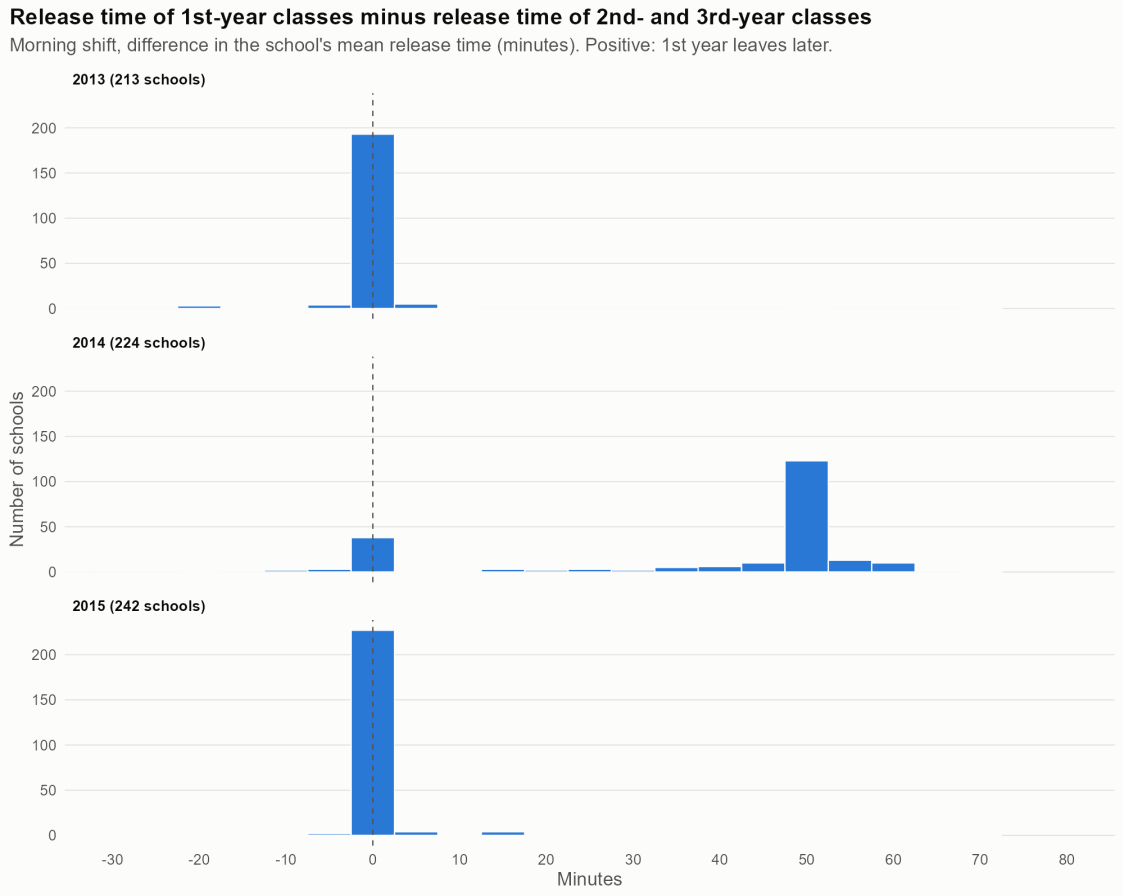


Figure 4: Difference, in each school, between the mean release time of 1st-year EM classes and the mean release time of 2nd- and 3rd-year EM classes, morning shift (classes starting before 12:00), in 2013, 2014 and 2015. Positive values: the 1st year leaves later. Sample state schools with both 1st-year and 2nd- or 3rd-year morning classes in the year. **Note:** the 19 schools that delayed 1st-year release by 50 minutes or more before 2014 (8 from 2011 to 2012 and 11 from 2012 to 2013) are excluded. The horizontal axis is cut at −30 and 80 minutes, which leaves out 3 schools in 2013 (differences of −120, −170 and −225 minutes).

$\overline{V}^{\text{shift}}_{\text{shared}}$ = average violence among 9th-grade cohorts in that shift whose release time coincides with high school's; $\overline{V}^{\text{shift}}_{\text{not shared}}$ = same, for cohorts whose release time does not coincide.

(d) TWFE, student level. Level. Student, across their own grade progression (8th → 9th). Student m is a within-cohort control whose shared-release status does not change from 8th to 9th grade.

$$\hat{\beta}_{DD} = \left(Y_{s,9\text{th}} - Y_{s,8\text{th}}\right) - \left(Y_{m,9\text{th}} - Y_{m,8\text{th}}\right) \tag{5}$$

$Y_{i,g}$ = violence outcome for student i in grade g; s = student whose shared-release status changes moving from 8th to 9th grade; m = student whose shared-release status stays the same across both grades.

5.5 Other effects

- Other schools near. Affect in other schools

- Problem: other schools can affect

6 Release times by EM grade

Fazer event sstudy: time of crime after school release time (X) Y probability of crime (student level) Problem: report of policy Problem: more people around; more policy

Mechanisms:

Do the DAG

Tres histograma: diferenca media do horario de saida das classes do primeiro ano em relacao ao segundo e terceiro ano. Escola X =

References

Billings, Stephen B., David J. Deming, and Stephen L. Ross, "Partners in Crime," *American Economic Journal: Applied Economics*, 2019, *11* (1), 126–150.

Jacob, Brian A. and Lars Lefgren, "Are Idle Hands the Devil's Workshop? Incapacitation, Concentration, and Juvenile Crime," *American Economic Review*, 2003, *93* (5), 1560–1577.

Lochner, Lance and Enrico Moretti, "The Effect of Education on Crime: Evidence from Prison Inmates, Arrests, and Self-Reports," *American Economic Review*, 2004, *94* (1), 155–189.

Luallen, Jeremy, "School's Out...Forever: A Study of Juvenile Crime, At-Risk Youths and Teacher Strikes," *Journal of Urban Economics*, 2006, *59* (1), 75–103.

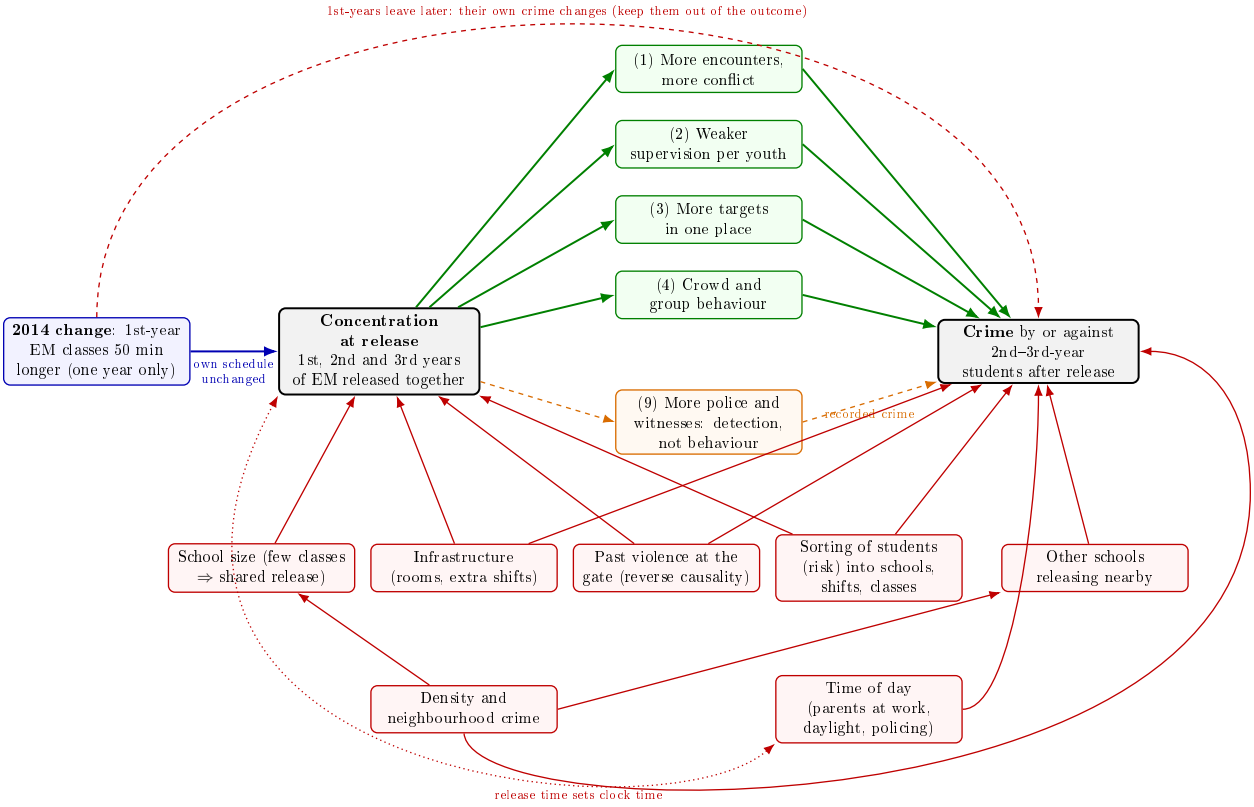


Figure 5: DAG for the effect of concentration at release on the crime of 2nd- and 3rd-year EM students. Green: concentration mechanisms (1)–(4), numbered as in the Mechanisms subsection. Orange dashed: detection (9), which raises recorded crime without changing behaviour. Red: confounders that open back doors between concentration and crime (school level, individual sorting, time of day). Blue: the 2014 change in 1st-year schedules, which moves concentration for 2nd- and 3rd-year students without changing their own schedule, so it closes the time-of-day and hours-in-school paths for them; the red dashed arc is its main threat. The peer mechanisms (5)–(8) are not in this graph: in this design the students exposed are older than those added to the crowd.

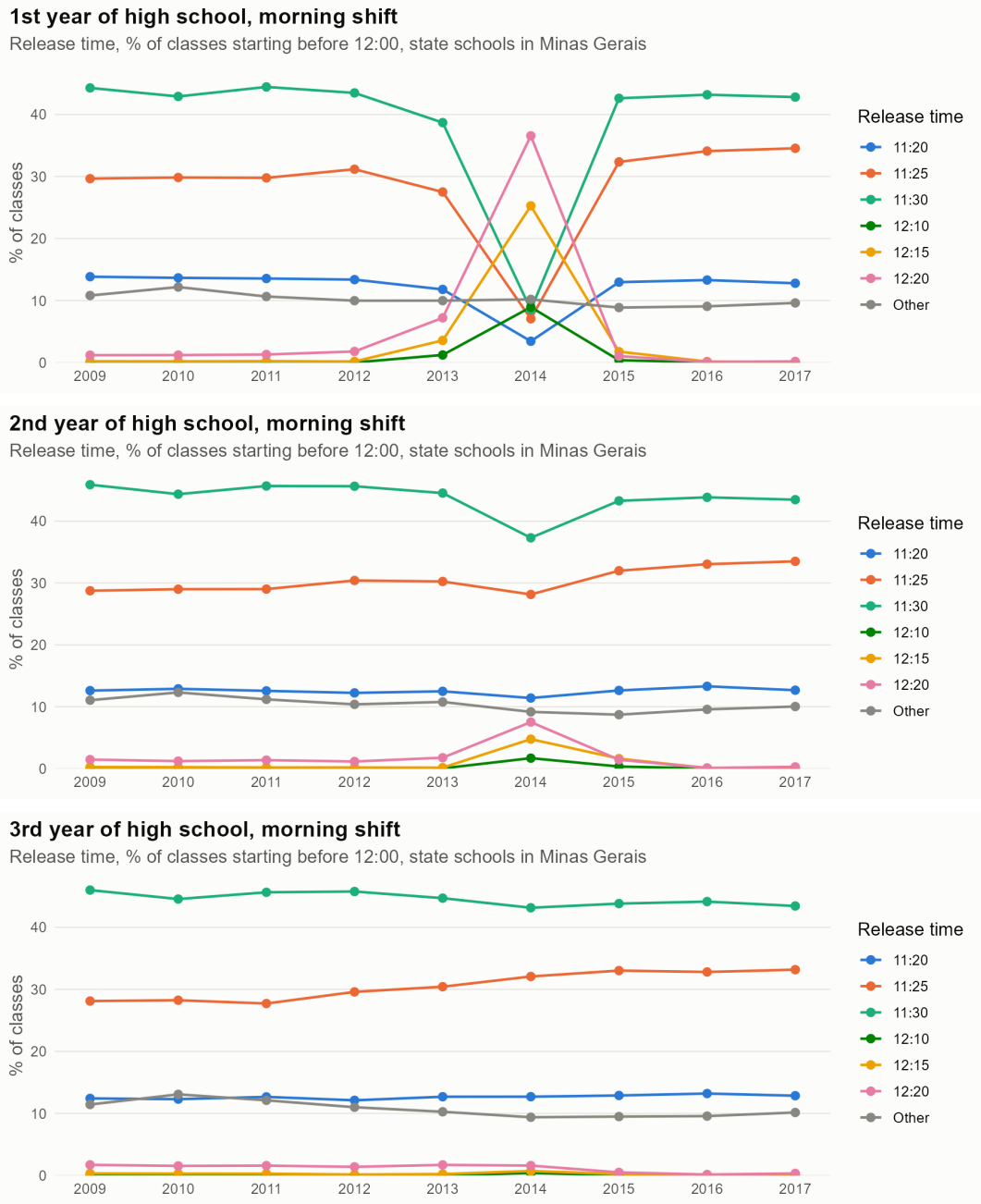


Figure 6: Release time of EM classes in the morning shift (classes starting before 12:00), by grade: 1st year (top), 2nd year (middle) and 3rd year (bottom). Share of the grade's classes released at each time, by year. All state schools in Minas Gerais. The same release times, colors and vertical axis are used in the three graphs; less frequent times are grouped as Other.

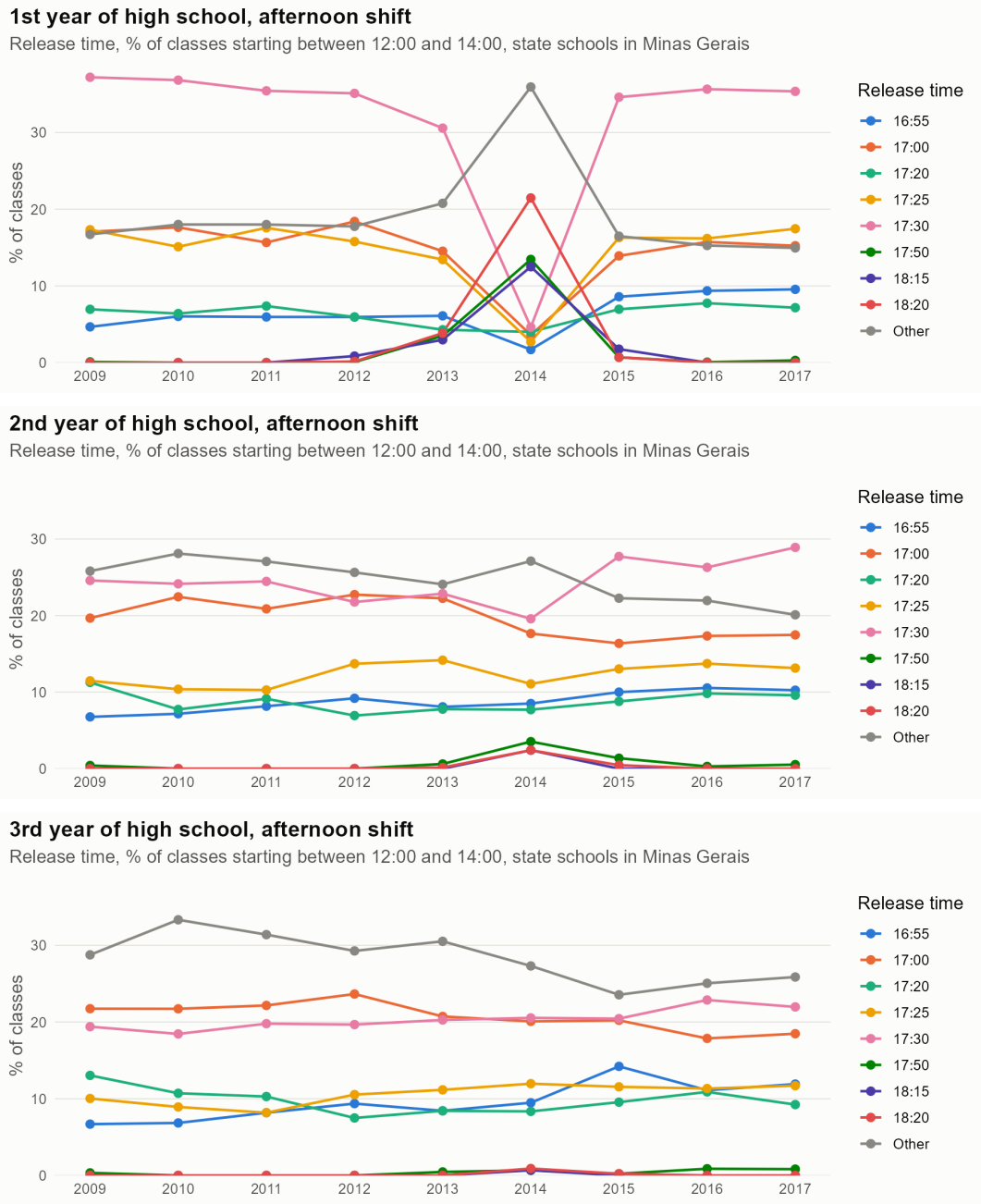


Figure 7: Release time of EM classes in the afternoon shift (classes starting between 12:00 and 14:00), by grade: 1st year (top), 2nd year (middle) and 3rd year (bottom). Share of the grade's classes released at each time, by year. All state schools in Minas Gerais. The same release times, colors and vertical axis are used in the three graphs; less frequent times are grouped as Other.